Applied Convex Optimization

TAMU ECEN-629

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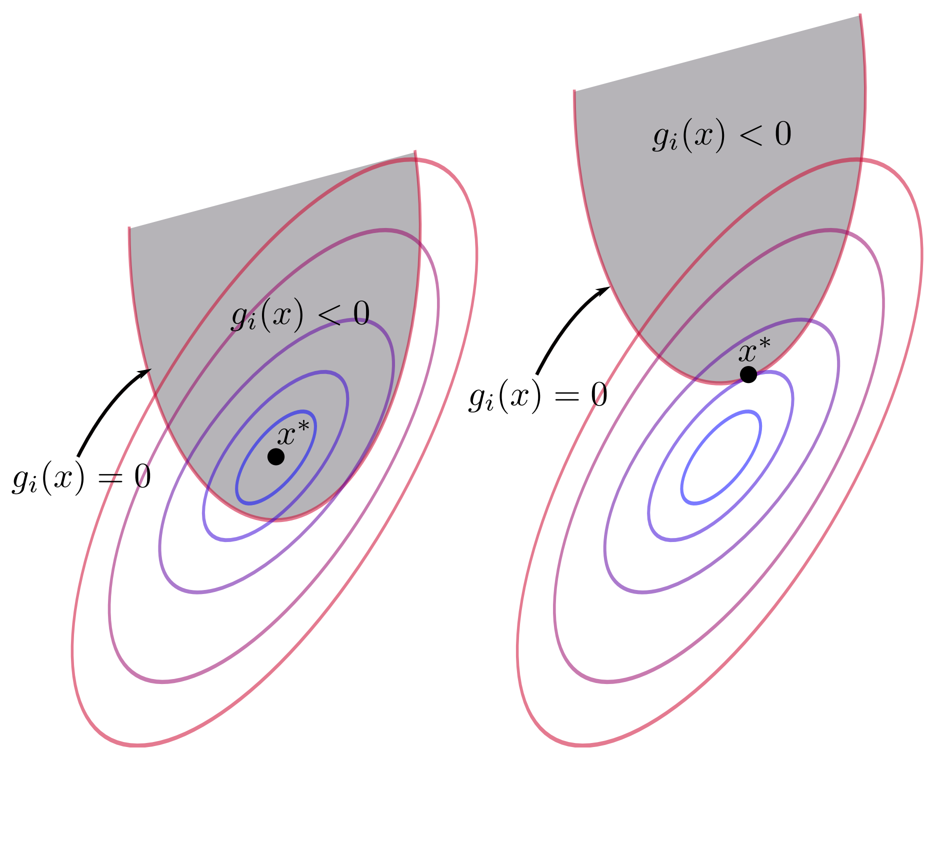
Fall 2025



Acknowledgements

These notes were compiled from the course materials for ECEN-629 (Applied Convex Optimization) taught by Dr. Chao Tian at Texas A&M University during the Fall 2025 semester.

Applied Convex Optimization (ECEN-629)

Instructor: Dr. Chao Tian
Semester: Fall 2025
Time: TTh 12:45pm - 2:00pm
Location: ETB 1003

Description: Introduction to convex optimization including convex set, convex functions, convex optimization problems, KKT conditions and duality, unconstrained optimization, and interior-point methods for constrained optimization; applications in information science, digital systems, networks and learning.

Contents

Chapter 1

Introduction

Optimization. We regularly encounter many optimizations in our daily lives. Intuitively, we think of optimization as finding the best way for *something*. In mathematics, optimization has a more formal meaning. It is defined rigorously with an objective function, a set of variables, and a set of constraints. A mathematical optimization problem has the following form:

$$\begin{aligned} &\text{minimize} && f_0(x) \\ &\text{subject to} && f_i(x) \le b_i, \quad i = 1, \ldots, m \end{aligned}$$

- $x = (x_1, \ldots, x_n)$
- $f_0 : \mathbb{R}^n \to \mathbb{R}$ is the objective function
- $f_i : \mathbb{R}^n \to \mathbb{R}$ are the constraint functions

The **optimal solution** $x^\star$, yields the minimal value of f_0 among all vectors x that satisfy the constraints. **optimal solution**

Generally, optimization problems are very difficult to solve. Methods for solving general optimization problems typically involve compromise, such as heightened time complexity or approximate solutions.

1.1 Optimization problem classes

There are a few problem classes that can be solved efficiently and reliably.

1.1.1 Least-squares problems

These problems are in the form

$$\text{minimize } ||Ax - b||_2^2$$

The analytical solution has form $x^\star = (A^TA)^{-1}A^Tb$. It is reliable, with computational time proportional to n^2k ($A \in \mathbb{R}^{k\times n}$) Least-squares problems are typically easy to recognize, and there exist a few techniques to increase flexibility (weights, regularization).

1.1.2 Linear programming problems

These problems are in the form

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \ldots, m \end{array}$$

There exists no analytical formula for solution. These problems have computational time proportional to $n^2 m$ if $m \geq n$. Typically these problems are not as easy to recognize as least-squares problems. Luckily, there are a few standard tricks that help convert problems into linear programs (ℓ_1)

1.1.3 Convex optimization problems

The primary focus of this course. These problems are in the form

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \ldots, m \end{array}$$

With these problems, the objective and constraint functions are convex:

$$f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$$

if $\alpha + \beta = 1$, $\alpha \geq 0$, $\beta \geq 0$.

Least-squares problems and linear programs are special cases of convex optimization problems. Computation time for convex optimization problems is roughly proportional to $\max\{n^3, n^2 m, F\}$, where F is the cost of evaluating f_i values and first and second derivatives. Many problems can be solved via convex optimization, though they are often difficult to recognize. However, there exist many tricks for transforming problems into convex form.

1.1.3.1 Convex example problem

Let's say we have m lamps illuminating n patches. Intensity I_k at patch k depends on lamp powers p_j:

$$I_k = \sum_{j=1}^{m} a_{kj} p_j \quad a_{kj}^{-2} \max\{\cos\theta_{kj},\, 0\}$$

We aim to achieve a desired illumination, I_{des}, with bounded lamp powers. Thus we state the problem as follows

$$\begin{array}{ll} \text{minimize} & \max_{k=1,\ldots,n} |\log I_k - \log I_{\text{des}}| \\ \text{subject to} & 0 \leq p_j \leq p_{\max}, \quad j = 1, \ldots, m \end{array}$$

We can solve this problem by converting it to a convex optimization problem. Now, the problem is stated as

$$\begin{array}{ll} \text{minimize} & f_0(p) = \max_{k=1,\ldots,n} h(I_k/I_{\text{des}}) \\ \text{subject to} & 0 \leq p_j \leq p_{\max}, \quad j = 1, \ldots, m \end{array}$$

where $h(u) = \max\{u, 1/u\}$.

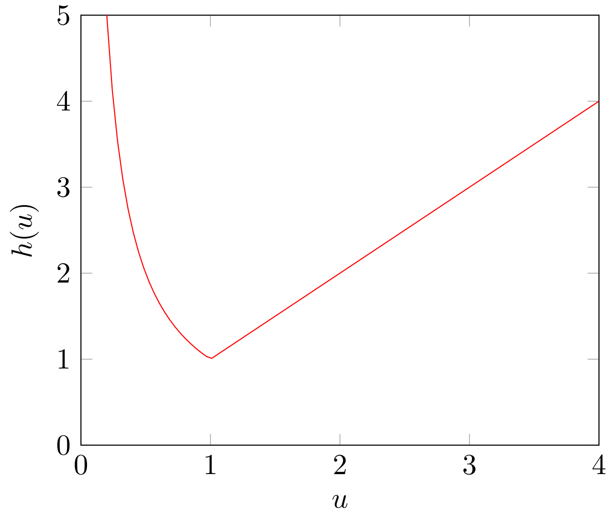


Figure 1.1: Illustration of the convex optimization problem ($h(u)$).

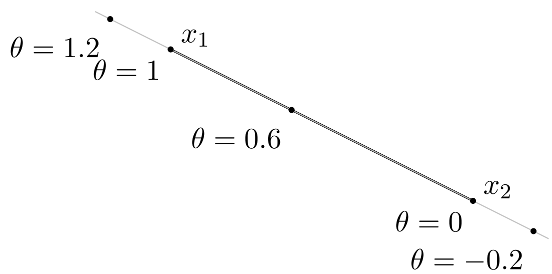
This transformation is valid because *the maximum of a convex function is convex.* These convex optimization problems will be the focus of this course.

Chapter 2

Convex sets

To start off, let's define a line segment between two points x_1 and x_2. The line segment can be expressed as

$$x = \theta x_1 + (1 - \theta)x_2, \quad (\theta \in \mathbb{R})$$



This figure will be useful to visualize the concepts of affine and convex sets. hey!!!!!

2.1 Affine set

We start this chapter by defining an *affine set*. Say we have some x_1 and x_2. An affine set contains all points on the line through x_1 and x_2.

$$x = \theta x_1 + (1 - \theta)x_2, \quad (\theta \in \mathbb{R})$$

Affine set: contains the line through any two distinct points in the set. And example of this sould be the solution set of linear equations $\{x \mid Ax + b\}$

2.2 Convex set

Convex set: line segment between x_1 and x_2: all points

$$x = \theta x_1 + (1 - \theta)x_2, \quad (0 \leq \theta \leq 1)$$

The convex set contains the line segment between *any* two points in set.

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \implies \theta x_1 + (1-\theta)x_2 \in C$$

Exercise 1

Let $C \subset \mathbb{R}^n$ be a convex set, x_1, x_2, $\ldots$, $x_k \in C$. Let θ_1, θ_2, $\ldots$, $\theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0$ and $\sum_{i=1}^{k} \theta_i = 1$. Show that

Show that $\theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_k x_k \in C$.

$$\begin{aligned} &\theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_k x_k = 1 \\ &= (\theta_1 + \theta_2)\left(\frac{\theta_1}{\theta_1 + \theta_2} x_1 + \frac{\theta_2}{\theta_1 + \theta_2} x_2\right) + \theta_3 x_3 \end{aligned}$$

Convex combination and convex hull

Convex combination of x_1, $\ldots$, x_k: any point x of form

$$x = \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_k x_k, \quad \theta_i \geq 0, \sum_{i=1}^{k} \theta_i = 1$$

Convex hull ($\operatorname{conv} S$): set of all convex combinations of points in S.